rebalancedNoise – R package

https://github.com/sdcTools/rebalancedNoise

Prototype implementation of the Swedish EZS perturbation method for magnitude tables

- https://unece.org/sites/default/files/2025-10/SDC2025_Sf_Sweden_Almberg_D.pdf
- currently under development in an R package.
- The method assigns each unit in the microdata two parameters:
  - a direction factor (−1 / +1)
  - a unit-specific perturbation factor (percentage), conceptually similar to record keys in the Cell-Key method.
- Initial perturbation step:
  - Perturbed cell values are calculated as the sum of perturbed micro-values derived from the original value, direction, and perturbation factor.
- Sensitive cells remain fixed, while non-sensitive cells undergo a rebalancing step:
  - Starting from the second-largest contributor, direction factors are adjusted to minimize the deviation from the original cell value.

Key methodological property

- The perturbation + rebalancing procedure is applied to **inner cells only**, with **aggregates computed from these perturbed cells**.
- This leads to **fully additive tables**, which is a major difference compared to the **Cell-Key method**, where additivity is generally not guaranteed.

Practical constraints

- **Consistency is guaranteed only within a predefined table structure.**
- In practice, this requires defining a **master table** from which all outputs are derived.
- Rebalancing requires **sufficiently populated cells**, so extremely sparse tables may reduce its effectiveness.

Comparison to Cell-Key perturbation

- **Advantages**
  - Fully additive tables.
  - Conceptually simple perturbation mechanism.
- **Trade-offs**
  - Output tables must be defined in advance.
  - Post-processing steps may break additivity, removing the main methodological advantage.

Open research questions and …

- Preservation **of logical relations between variables** when perturbing variables independently.
- Possible solution:
  - perform rebalancing on a **main variable** and reuse resulting direction factors for other variables.
- Interaction with **dominance rules** for sensitive cell detection still needs further testing.

… next steps

- Extensive **simulation and testing** to evaluate disclosure risk and data utility.
- **Release** in github.com/sdcTools/rebalancedNoise Repo to encourage external feedback and collaboration.